**What is JDK ?**

- JDK stands for Java Development Kit.
- JDK is the Java software. It provides
    - Development tools
    - Runtime environment
    - Predefined Library (API = Application Programming Interface) for developing, compiling and executing Java programs

- Development tools means small software for compiling java program, reading byte code, generating documentation, preparing jar file, etc.. Operations performing software are called development tools.

- Runtime environment means the platform for running java program. The JVM is called runtime environment or Java platform for executing java programs.

- Java library means the predefined classes provided by SUN(Oracle) as part of JDK for reusing and developing our new programs

- In Java we will calls Library as API (Application Programming Interface)

- Hence we can say by installing JDK software, we will get Compiler, JVM , API for developing, compiling and executing Java programs.

**Downloading and Installing JDK-22 Software**

1. Open Browser
2. Search jdk 22 downloads
3. Click Java Downloads
4. Scroll down -> Click Windows
5. Click jdk.exe file
6. Downloading will be start, wait

7. After downloading is completed
8. Click on jdk.exe file on browser
9. Click "Yes" to give permission to start installing
10. Click "Next" in the first Window
11. Click "Change" in the next window
12. And make installing path as "C:\jdk-22\"

13. Click Ok -> Next -> Close
14. JDK installation is completed in the folder "C:\jdk-22"

**JDK Software folder Structure**

- JDK Software binary and library files are organised in several folder as below
  - C : \ jdk - 22

```
| - bin
        1. javac.exe          (Compiler)
        2. java.exe           (JVM)
        3. javap.exe          (Java profiler -> byte code reader)
        4. javadoc.exe        (Java document generator)
        5. jar.exe
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       -------
| - lib
   - jrt-fs.jar      (*.class files)
   - src.zip         (*.java files)
|- jmods        (*.modules)
      |  - java.base.mod
      |  - java.desktop.mod
      |  - java.sql.mod
      |  ----
      |  ----
|- conf          ( *.configation files)
|- include        *.h (header files)
```

**Steps to Develop a Java Program**

- A java program development, compilation and execution comes under 7 steps
  1. Open Notepad
  2. Type Java Program
  3. Save this program with .java extension
  4. Open cmd prompt software
  5. Change drive and directory path in cmd prompt to our project drive and directory in which program is saved
  6. Compile Java program by using "javac" tool as

     javac <filename>

     For example : javac Add.java
     | -> Add.class

Change drive in cmd
D:

7. Execute Java byte code by using "java" tool as

```
java <classname>
```

For example :

```
java Add
|
| -> Output
```